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Solution E2: Kinematics

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Task 1: 3R Robot Kinematics

For the 3R manipulator shown in Figure 1, solve the following tasks:

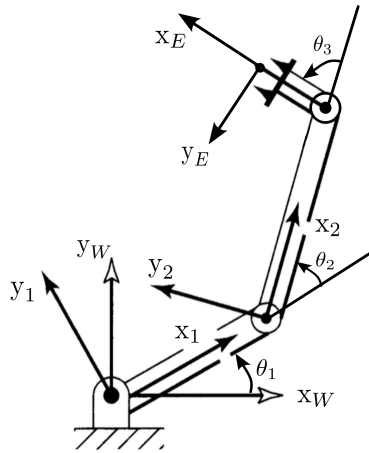


Figure 1: Planar 3R manipulator with parallel joint axes. The configuration is defined by joint angles $\theta_1, \theta_2, \theta_3$ and link lengths l_1, l_2, l_3 . A world frame $\underline{F}_W$ is attached at the base. An end-effector frame $\underline{F}_E$ is attached to the gripper tip point as shown.

Hint: to simplify the presentation of the Jacobian matrix, we adopt the following shorthand notation for the trigonometric functions: $s_i = \sin(\theta_i)$, $c_i = \cos(\theta_i)$, $s_{ij} = \sin(\theta_i + \theta_j)$, $c_{ij} = \cos(\theta_i + \theta_j)$ etc. in some part of this solution.

- i) Let, for notational brevity, $[t_x, t_y]^T := {}_W\mathbf{t}_{WE}$ and ϕ_E the orientation of the end-effector (angle from x_W to x_E). Compute the forward kinematics, i.e., the end-effector pose $[t_x, t_y, \phi_E]^T$ expressed in $\underline{F}_W$ as a function of the joint states $\boldsymbol{\theta} = [\theta_1, \theta_2, \theta_3]^T$.

A minimal planar pose description of the end-effector in $\underline{F}_W$ is given by

$$\begin{pmatrix} t_x \\ t_y \\ \phi_E \end{pmatrix} = \begin{pmatrix} l_1 \cos \theta_1 + l_2 \cos(\theta_1 + \theta_2) + l_3 \cos(\theta_1 + \theta_2 + \theta_3) \\ l_1 \sin \theta_1 + l_2 \sin(\theta_1 + \theta_2) + l_3 \sin(\theta_1 + \theta_2 + \theta_3) \\ \theta_1 + \theta_2 + \theta_3 \end{pmatrix}. \quad (1)$$

- ii) The Jacobian $\mathbf{J}(\boldsymbol{\theta})$ of the manipulator expressed in $\underline{F}_W$ is defined as

$$\mathbf{J} := \partial[t_x, t_y, \phi_E]^T / \partial \boldsymbol{\theta}$$

and it can be used to map between joint velocities and end-effector twist as

$$[\dot{t}_x, \dot{t}_y, \dot{\phi}_E]^T = \mathbf{J}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}$$

Derive the analytical expression for $\mathbf{J}(\boldsymbol{\theta})$.

Here, we simply need to compute the Jacobian of the position description specified above:

$$\mathbf{J}(\boldsymbol{\theta}) = \begin{pmatrix} \frac{\partial p_1}{\partial \theta_1} & \frac{\partial p_1}{\partial \theta_2} & \frac{\partial p_1}{\partial \theta_3} \\ \frac{\partial p_2}{\partial \theta_1} & \frac{\partial p_2}{\partial \theta_2} & \frac{\partial p_2}{\partial \theta_3} \\ \frac{\partial p_3}{\partial \theta_1} & \frac{\partial p_3}{\partial \theta_2} & \frac{\partial p_3}{\partial \theta_3} \end{pmatrix} \quad (2)$$

The partial derivatives can be evaluated quite simply, and the Jacobian is then:

$$\mathbf{J}(\boldsymbol{\theta}) = \begin{pmatrix} -l_3 s_{123} - l_2 s_{12} - l_1 s_1 & -l_3 s_{123} - l_2 s_{12} & -l_3 s_{123} \\ l_3 c_{123} + l_2 c_{12} + l_1 c_1 & l_3 c_{123} + l_2 c_{12} & l_3 c_{123} \\ 1 & 1 & 1 \end{pmatrix} \quad (3)$$

- iii) Determine the singular configurations of the manipulator, i.e. $\boldsymbol{\theta}$ s.t. $\det(\mathbf{J}(\boldsymbol{\theta})) = 0$. Can you provide an intuitive explanation of the singularities and determine which degrees of freedom are lost at these configurations?

Singularity means the robot loses one or more degrees of instantaneous motion. Formally:

$$\text{rank}(\mathbf{J}(\boldsymbol{\theta})) < \max_{\boldsymbol{\theta}'} \text{rank}(\mathbf{J}(\boldsymbol{\theta}')) \quad (4)$$

and $\det(\mathbf{J}(\boldsymbol{\theta})) = 0$. For $\mathbf{J} \in \mathbb{R}^{3 \times 3}$,

$$\begin{aligned} \det(\mathbf{J}(\boldsymbol{\theta})) &= \begin{vmatrix} -l_1 s_1 - l_2 s_{12} - l_3 s_{123} & -l_2 s_{12} - l_3 s_{123} & -l_3 s_{123} \\ l_1 c_1 + l_2 c_{12} + l_3 c_{123} & l_2 c_{12} + l_3 c_{123} & l_3 c_{123} \\ 1 & 1 & 1 \end{vmatrix} \\ &= \begin{vmatrix} -l_1 s_1 & -l_2 s_{12} & -l_3 s_{123} \\ l_1 c_1 & l_2 c_{12} & l_3 c_{123} \\ 0 & 0 & 1 \end{vmatrix} \\ &= -l_1 l_2 s_1 c_{12} + l_1 l_2 s_{12} c_1 \end{aligned}$$

where we simplified the determinant calculations through linear combinations of the columns (preserves rank). We then solve for:

$$-l_1 l_2 s_1 c_{12} + l_1 l_2 s_{12} c_1 = 0 \implies s_{12} c_1 - s_1 c_{12} = 0 \quad (5)$$

We use the trigonometric identities

$$\sin x \cos y = \frac{1}{2}(\sin(x - y) + \sin(x + y)) \quad (6)$$

to obtain

$$\begin{aligned} s_{12} c_1 &= \frac{1}{2}(\sin(\theta_2) + \sin(2\theta_1 + \theta_2)) \\ s_1 c_{12} &= \frac{1}{2}(\sin(-\theta_2) + \sin(2\theta_1 + \theta_2)) \end{aligned}$$

from which it follows that

$$\sin \theta_2 = 0$$

must hold for a singular configuration. Thus, the solution is $\theta_2 \in \{0^\circ, 180^\circ\}$.

Figure 2 shows two intuitive singular positions, for $\theta_2 = \theta_3 = 0$ and $\theta_2 = 0$. The first row shows workspace boundary, where the robot is full stretched and it is clear that movement in the x_E -direction is not possible. We can also verify this by looking at the Jacobian. If we set $l_1 = l_2 = l_3 = 1$, the Jacobian at $(0^\circ, 0^\circ, 0^\circ)$ is:

$$\mathbf{J}(0^\circ, 0^\circ, 0^\circ) = \begin{pmatrix} 0 & 0 & 0 \\ 3 & 2 & 1 \\ 1 & 1 & 1 \end{pmatrix} \quad (7)$$

We see that the first row is all zero, which means $\dot{x} = 0$ for any $\dot{\theta}$. At this configuration, all joint axes are aligned and all links are collinear along the x_E -axis. The velocity of each joint is perpendicular to its link, i.e. purely in y_E -axis-direction.

The second workspace interior configuration is harder to determine which movement is impossible. Let's look at a concrete example of the Jacobian for that configuration, with, let's say, $\theta_1 = 90^\circ, \theta_2 = 0^\circ, \theta_3 = -90^\circ$, and all lengths are set to 1 equally. The Jacobian looks like this:

$$\mathbf{J}(90^\circ, 0^\circ, -90^\circ) = \begin{pmatrix} -2 & -1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

So we see that the last two rows of the Jacobian are linearly dependent. If we imagine multiplying any vector $\dot{\theta}$ with that matrix as in eq. 8, we see that the angular velocity $\dot{\phi}_E$ and the y velocity $\dot{t}_y$ will be coupled directly:

$$\begin{pmatrix} \dot{t}_x \\ \dot{t}_y \\ \dot{\phi}_E \end{pmatrix} = \mathbf{J}(\theta) \cdot \begin{pmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \end{pmatrix} \quad (8)$$

$$\begin{pmatrix} \dot{t}_x \\ \dot{t}_y \\ \dot{\phi}_E \end{pmatrix} = \begin{pmatrix} -2\dot{\theta}_1 - \dot{\theta}_2 \\ \dot{\theta}_1 + \dot{\theta}_2 + \dot{\theta}_3 \\ \dot{\theta}_1 + \dot{\theta}_2 + \dot{\theta}_3 \end{pmatrix} \quad (9)$$

meaning that we cannot independently choose $\dot{\phi}_E$ and $\dot{t}_y$. E.g. producing pure $\dot{\phi}_E$ without producing $\dot{t}_y$ (and vice versa) is impossible.

The configurations shown here are only representative examples of singularities for the planar 3R manipulator. In fact, singularities occur whenever adjacent links become collinear (i.e., when $\theta_2 = 0$ or $\theta_2 = \pi$).

Task 2 (Advanced): Isotropic Points

Besides singularities, a manipulator may exhibit other special configurations, such as isotropic configurations. These are characterized by the Jacobian matrix having orthogonal columns of equal norm, i.e., $\mathbf{J}^T \mathbf{J} = \delta \mathbf{I}$, $\delta > 0$. Consider the 2R manipulator shown in Figure 3. Its Jacobian in end-effector frame $\underline{\mathcal{F}}_E$ (with respect to gripper position only, ignoring orientation) is given as:

$$\mathbf{J}(\theta) = \begin{pmatrix} l_1 \sin \theta_2 & 0 \\ l_1 \cos \theta_2 + l_2 & l_2 \end{pmatrix}$$

- i) Determine the manipulator's isotropic points (treat both θ_i and l_i as free variables) and draw the manipulator in the corresponding configuration(s).

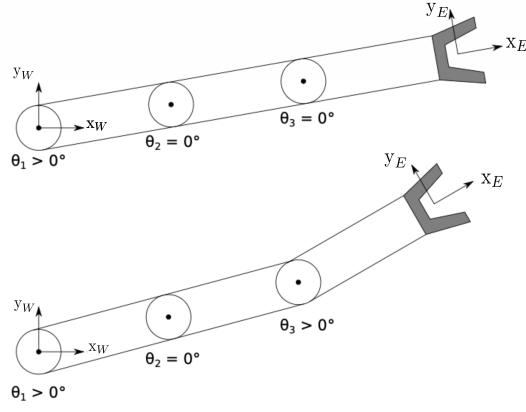


Figure 2: Singular configurations. First row: at workspace boundary. Second row: workspace interior.

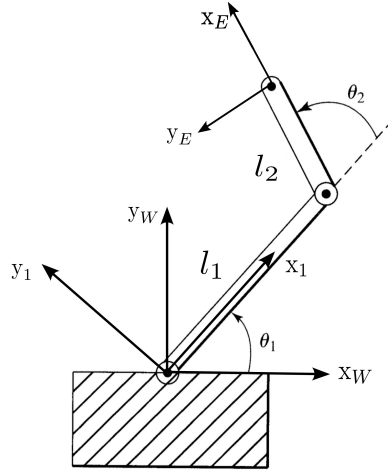


Figure 3: 2R Robot configured using joint parameters θ_1, θ_2 and link lengths l_1, l_2 .

Since the column vectors have to be orthogonal, we have:

$$\begin{aligned} (l_1 s_2, l_1 c_2 + l_2) \cdot (0, l_2)^T &= 0 \quad \Leftrightarrow \\ l_2(l_1 c_2 + l_2) &= 0 \quad \Leftrightarrow \\ c_2 &= -\frac{l_2}{l_1} \end{aligned}$$

Also, the vectors have to be of equal length, so we have:

$$\begin{aligned} l_1^2 s_2^2 + (l_1 c_2 + l_2)^2 &= l_2^2 \quad \Leftrightarrow \\ l_1^2 s_2^2 + l_1^2 c_2^2 + 2l_1 c_2 l_2 + l_2^2 &= l_2^2 \quad \Leftrightarrow \\ l_1^2 + 2l_1 c_2 l_2 &= 0 \quad \Leftrightarrow \\ c_2 &= -\frac{l_1}{2l_2} \end{aligned}$$

Combining both constraints, we conclude:

$$\begin{aligned} -\frac{l_1}{2l_2} &= -\frac{l_2}{l_1} \quad \Leftrightarrow \\ l_1^2 &= 2l_2^2 \quad \Rightarrow \\ l_1 &= \sqrt{2}l_2 \end{aligned}$$

This yields the solution $c_2 = \frac{-1}{\sqrt{2}} \Rightarrow \theta_2 = \pm 135^\circ$. The Jacobian in this position looks like this:

$$\mathbf{J} = \begin{pmatrix} \frac{l_1}{\sqrt{2}} & 0 \\ -\frac{l_1}{\sqrt{2}} + \frac{l_1}{\sqrt{2}} & l_2 \end{pmatrix} = \begin{pmatrix} l_2 & 0 \\ 0 & l_2 \end{pmatrix}$$

ii) Provide an intuitive interpretation of the special role that the isotropic configuration plays.

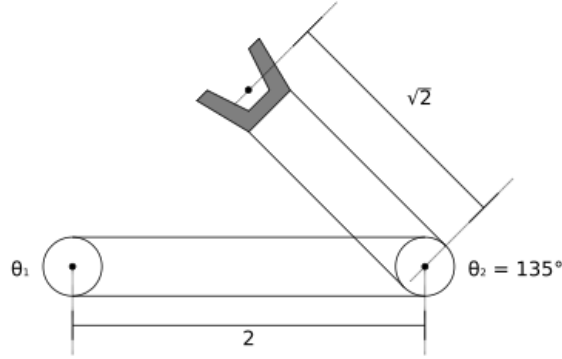


Figure 4: Robot from task 2 in isotropic configuration.

From the Jacobian of this specific configuration and link-length choice, one can see now that $\dot{\theta}_1$ and $\dot{\theta}_2$ map directly to a velocity in x_E and y_E direction, with same velocity amplification. This means the robot moves equally well in both x_E and y_E direction and has so-called best dexterity. The robot in this configuration is shown in Figure 4.

Task 3 (Advanced): Kinematics Computation in Python

In this task, you will implement forward kinematics (FK), Jacobian computation, and inverse kinematics (IK) for a mobile manipulator (Tiago) using a modern kinematics library (e.g., Pinocchio).

By completing this task, you should be able to:

- Understand the robot kinematic model derived from a Universal Robot Description File (URDF).
- Compute forward kinematics and extract end-effector poses.
- Compute the geometric (position) Jacobian of the end-effector.
- Implement a numerical inverse kinematics (IK) solver.

You will begin with model inspection and visualization, then move to Jacobian-based kinematics computation, and finally implement a numerical inverse kinematics solver. Follow the step-by-step instructions provided in the Jupyter notebook.

For a reference implementation of this task, please check `e02_sol.ipynb`.

i) Model Loading and Visualization

- Load the robot model from its URDF file.
- Inspect the kinematic structure (joints, frames, DoFs).

- Visualize the robot home configuration.

ii) Numerical Inverse Kinematics

The mapping from joint states to end-effector pose is often non-linear and an analytical inverse may not exist (cf. overactuated systems, i.e. if the degrees of freedom provided by the joints are larger than the end-effector DoFs. Think of yourself being able to reach for an object with one hand while still maintaining a lot of freedom in how exactly you pose your body for this task).

Nevertheless, we can solve for joint states given a desired end-effector pose using an iterative solving scheme. Given a desired end-effector position¹ ${}_W\mathbf{t}_{WE,goal}$, we leverage the associated Jacobian $\mathbf{J}(\boldsymbol{\theta})$: $d({}_W\mathbf{t}_{WE})/dt = \mathbf{J}(\boldsymbol{\theta})\dot{\boldsymbol{\theta}}$ – which is easy to obtain and (pseudo) invert – for relating small increments Δ in end-effector position $\Delta({}_W\mathbf{t}_{WE})$ and, correspondingly, the individual joints states $\Delta\boldsymbol{\theta}$. Using the Jacobian as linearization of the nonlinear forward kinematics $\Delta({}_W\mathbf{t}_{WE}) \approx \mathbf{J}(\boldsymbol{\theta})\Delta\boldsymbol{\theta}$, we can solve for $\Delta\boldsymbol{\theta}$ using the pseudo-inverse $\mathbf{J}^\dagger(\boldsymbol{\theta})$

$$\Delta\boldsymbol{\theta} = \mathbf{J}(\boldsymbol{\theta})^\dagger \Delta({}_W\mathbf{t}_{WE})$$

with

$$\mathbf{J}(\boldsymbol{\theta})^\dagger = \mathbf{J}(\boldsymbol{\theta})^T (\mathbf{J}(\boldsymbol{\theta})\mathbf{J}(\boldsymbol{\theta})^T)^{-1}$$

and

$$\Delta({}_W\mathbf{t}_{WE}) = k_p({}_W\mathbf{t}_{WE,goal} - {}_W\mathbf{t}_{WE})$$

designed to take incremental steps towards the goal. Update

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \Delta\boldsymbol{\theta}$$

and iterate until the position error norm falls below a predefined threshold.

- $\mathbf{J}$ is the end-effector Jacobian,
- $\mathbf{J}^\dagger$ is its Moore–Penrose pseudo-inverse,
- k_p is a proportional gain,
- $\Delta\boldsymbol{\theta}$ is the joint increment.

¹Analog reasoning for orientation. For the scope of this exercise, however, we consider only the position.